Import Dependencies

```
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# FINAL PROJECT WEEK 2 - CS30
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# Install dependencies
!pip install kagglehub

import os
import random
import numpy as np
import matplotlib.pyplot as plt
from PIL import Image

import torch
import torch.nn as nn
from torch.utils.data import random_split, DataLoader
from torchvision import datasets, transforms, models
```

```
Requirement already satisfied: kagglehub in /usr/local/lib/python3.12/dist-packages (1.0.0)
Requirement already satisfied: kagglesdk<1.0,>=0.1.14 in /usr/local/lib/python3.12/dist-packages (from kag
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Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from kagglehub) (4.67.3)
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Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from request
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from request
```

1. Download Dataset from Kaggle

```
import kagglehub

path = kagglehub.dataset_download("emmarex/plantdisease")
dataset_path = os.path.join(path, "PlantVillage")

print("Dataset path:", dataset_path)
```

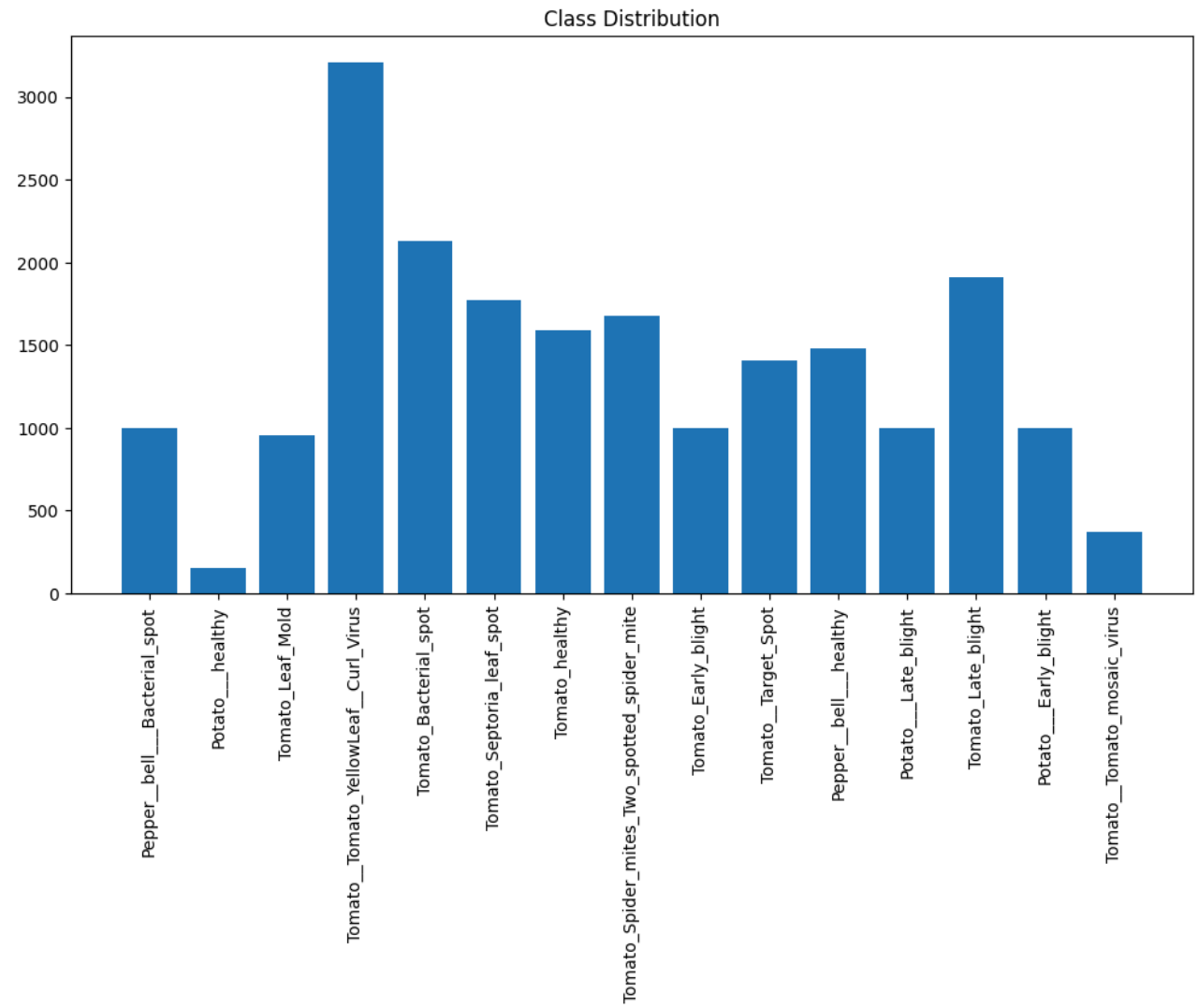
```
Using Colab cache for faster access to the 'plantdisease' dataset.
Dataset path: /kaggle/input/plantdisease/PlantVillage
```

2. EDA - CLASS DISTRIBUTION

```
classes = [cls for cls in os.listdir(dataset_path)
           if os.path.isdir(os.path.join(dataset_path, cls))]

class_counts = {}

for cls in classes:
    class_path = os.path.join(dataset_path, cls)
    images = [img for img in os.listdir(class_path)
```

```
            if img.lower().endswith((".jpg",".png",".jpeg"))]
    class_counts[cls] = len(images)

print("Total classes:", len(classes))

# Bar chart
plt.figure(figsize=(12,6))
plt.bar(class_counts.keys(), class_counts.values())
plt.xticks(rotation=90)
plt.title("Class Distribution")
plt.show()
```

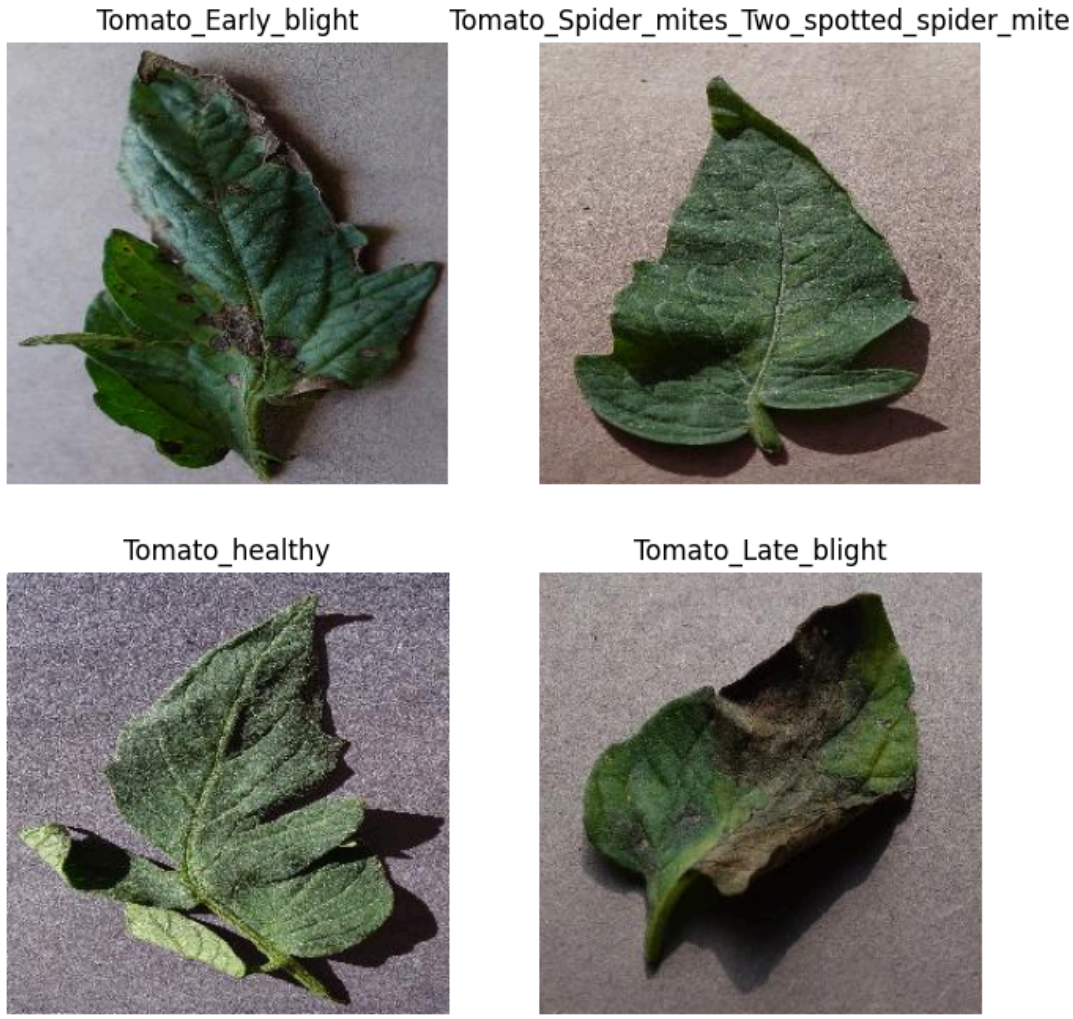
```
Total classes: 15
```



3. Sample Images

```
plt.figure(figsize=(8,8))

selected = random.sample(classes, 4)

for i, cls in enumerate(selected):
    class_path = os.path.join(dataset_path, cls)
    img_name = random.choice(os.listdir(class_path))
    img = Image.open(os.path.join(class_path, img_name))
```

```
    plt.subplot(2,2,i+1)
    plt.imshow(img)
    plt.title(cls)
    plt.axis("off")

plt.show()
```



4. Preprocessing

```
transform = transforms.Compose([
    transforms.Resize((224,224)),
    transforms.ToTensor(),
    transforms.Normalize(
        [0.485,0.456,0.406],
        [0.229,0.224,0.225]
    )
])

dataset = datasets.ImageFolder(dataset_path, transform=transform)
```

5. TRAIN / VAL / TEST Data Split

```
train_size = int(0.7 * len(dataset))
val_size = int(0.15 * len(dataset))
test_size = len(dataset) - train_size - val_size

train_data, val_data, test_data = random_split(
    dataset, [train_size, val_size, test_size]
)

train_loader = DataLoader(train_data, batch_size=32, shuffle=True)
val_loader = DataLoader(val_data, batch_size=32)
test_loader = DataLoader(test_data, batch_size=32)

print("Train:", len(train_data))
print("Val:", len(val_data))
print("Test:", len(test_data))
```

```
Train: 14446
Val: 3095
Test: 3097
```

6. SIMPLE ML - BASELINE MODEL

```
print("\nRunning baseline model...")

from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

X = []
y = []

# Randomly sample images from dataset
indices = random.sample(range(len(dataset)), 500)  # increase sample size

for i in indices:
    img, label = dataset[i]
    X.append(img.numpy().flatten())
    y.append(label)

X = np.array(X)
y = np.array(y)

print("Unique classes in sample:", np.unique(y))

model_lr = LogisticRegression(max_iter=1000)
model_lr.fit(X, y)

preds = model_lr.predict(X)

print("Baseline Accuracy:", accuracy_score(y, preds))
```

```
Running baseline model...
Unique classes in sample: [ 0  1  2  3  4  5  6  7  8  9 10 11 12 13 14]
Baseline Accuracy: 1.0
```

7. MobileNetV3 Model (Initial)

```
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
```

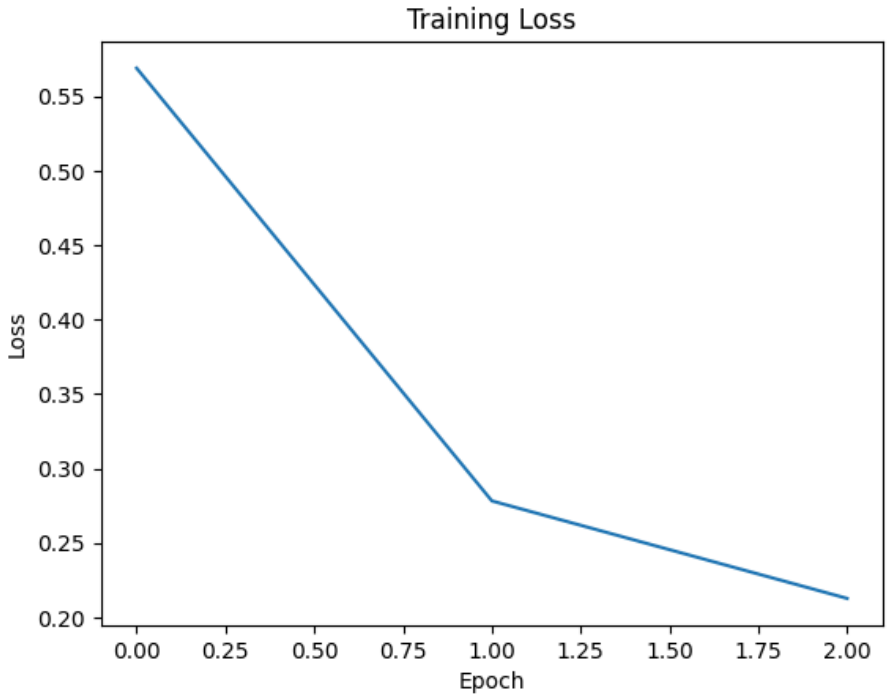
```
model = models.mobilenet_v3_small(pretrained=True)

# Replace classifier
model.classifier[3] = nn.Linear(
    model.classifier[3].in_features,
    len(classes)
)

# Freeze feature layers
for param in model.features.parameters():
    param.requires_grad = False

model = model.to(device)

criterion = nn.CrossEntropyLoss()
optimizer = torch.optim.Adam(model.parameters(), lr=0.001)
```

```
/usr/local/lib/python3.12/dist-packages/torchvision/models/_utils.py:208: UserWarning: The parameter 'pret
  warnings.warn(
/usr/local/lib/python3.12/dist-packages/torchvision/models/_utils.py:223: UserWarning: Arguments other tha
  warnings.warn(msg)
Downloading: "https://download.pytorch.org/models/mobilenet_v3_small-047dcff4.pth" to /root/.cache/torch/h
100%|██████████| 9.83M/9.83M [00:00<00:00, 76.6MB/s]
```

8. Training Loop

```
print("\nTraining MobileNetV3...")

epochs = 3
train_losses = []

for epoch in range(epochs):
    model.train()
    running_loss = 0

    for images, labels in train_loader:
        images, labels = images.to(device), labels.to(device)

        outputs = model(images)
        loss = criterion(outputs, labels)

        optimizer.zero_grad()
        loss.backward()
        optimizer.step()

        running_loss += loss.item()

    avg_loss = running_loss / len(train_loader)
    train_losses.append(avg_loss)

    print(f"Epoch {epoch+1}, Loss: {avg_loss:.4f}")

# Plot loss
plt.plot(train_losses)
plt.title("Training Loss")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.show()
```

```
Training MobileNetV3...
Epoch 1, Loss: 0.5689
Epoch 2, Loss: 0.2781
Epoch 3, Loss: 0.2126
```



9. Validation Accuracy

```
model.eval()

correct = 0
total = 0

with torch.no_grad():
    for images, labels in val_loader:
        images, labels = images.to(device), labels.to(device)

        outputs = model(images)
        _, predicted = torch.max(outputs, 1)

        total += labels.size(0)
        correct += (predicted == labels).sum().item()

accuracy = 100 * correct / total
print("Validation Accuracy:", accuracy)
```

```
Validation Accuracy: 93.50565428109854
```

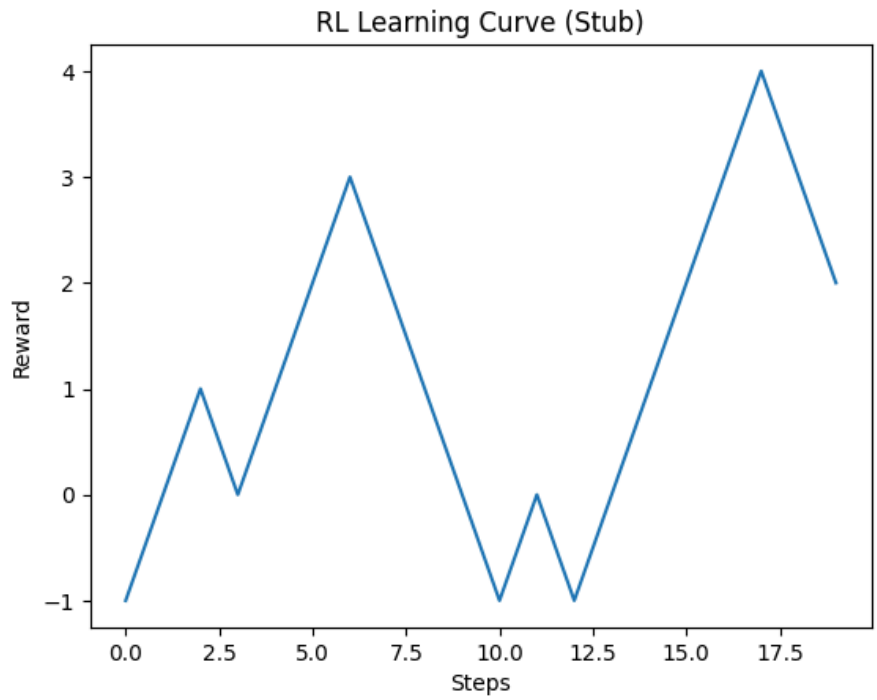
10. Simple NLP Component

```
def explain_prediction(class_name):
    return f"The model predicts {class_name} based on leaf patterns."

print(explain_prediction(classes[0]))
```

```
The model predicts Pepper__bell___Bacterial_spot based on leaf patterns.
```

11. RL Agent

```
class RLAgent:
    def __init__(self):
        self.reward = 0
        self.history = []

    def update(self, correct):
        if correct:
            self.reward += 1
        else:
            self.reward -= 1
        self.history.append(self.reward)

agent = RLAgent()

# simulate rewards
for i in range(20):
    agent.update(random.choice([True, False]))

plt.plot(agent.history)
plt.title("RL Learning Curve (Stub)")
plt.xlabel("Steps")
plt.ylabel("Reward")
plt.show()
```



12. Results

```
results = {
    "baseline_accuracy": float(accuracy_score(y, preds)),
    "mobilenet_val_accuracy": float(accuracy),
    "epochs": epochs
```

```
}

print("\nFinal Results:", results)
```

```
Final Results: {'baseline_accuracy': 1.0, 'mobilenet_val_accuracy': 93.50565428109854, 'epochs': 3}
```